Functional Description

Diesel engine-generator sets

with MTU 10/12 V 1600 Gx0 engines

Built in Europe

MS13021/04E

System designation	Engine model	Application group
DP00450D5S	10V1600G10F	3B, Continuous operation, variable
DP00500D5S	10V1600G20F	3B, Continuous operation, variable
DS00500D5S	10V1600G70F	3D, Short-term operation, fuel stop
DS00550D5S	10V1600G80F	3D, Short-term operation, fuel stop
DP00590D5S	12V1600G10F	3B, Continuous operation, variable
DP00650D5S	12V1600G20F	3B, Continuous operation, variable
DS00650D5S	12V1600G70F	3D, Short-term operation, fuel stop
DS00715D5S	12V1600G80F	3D, Short-term operation, fuel stop

Table 1: Applicability

Printed in Germany

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Table of Contents

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1 Engine-Generator Set

1.1 Engine-generator set – General information

The engine-generator set consists mainly of a diesel engine and a generator mounted on a common base frame. Upon demand, the engine will start and drives the generator to produce electrical power.

Engine-generator sets are available in two application groups: prime power and standby power.

Prime power

Prime power is the need for power, continuously and for long periods of time. It is often used in remote or developing areas where there is no utility service, where available service is very expensive or unreliable, or where customers choose to self-generate their primary power supply.

Prime power	Application group 3B
Mode of operation	Continuous operation with variable load
Rating definition	10 % overload capacity (ICXN)
Load factor	< 75 %
Operating hours	unrestricted

Standby power

Standby power systems are a type of system to provide backup resources in a crisis or when regular systems (e.g. power grid) fail. They find uses in a wide variety of settings from residential homes to hospitals, airports, scientific laboratories and data centers.

Standby power	Application group 3D
Mode of operation	Standby operation with variable load
Rating definition	Fuel stop power (IFN)
Load factor	< 85 %
Operating hours	max. 500 hours per year

Model designation of engine-generator sets

Key to the engine-generator set model designation	
D	Type: Diesel engine-generator set
P / S	Application: • P – Prime power (Application group 3B) • S – Standby power (Application group 3D)
00650	Performance: Rated power in kVA
D	Fuel type: Diesel
5	Frequency: 50 Hz
S	Project type: Standard product

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Benefits

- Wide range of standardized engine-generator sets to meet customer requirements with regard to power, emission and other characteristics
- Possibility to choose from different component designs (e. g. fuel prefilters) and to select options (e. g. fuel cooler)
- Latest diesel engine technology
- State-of-the-art main components for high efficiency and long service life

1.2 Engine-generator set – Standard scope of supply

The figures show an overview of the standard components on an engine-generator set with a 12 V 1600 Gx0 engine. The figures are also valid for an engine-generator set with a 10 V 1600 Gx0 engine.

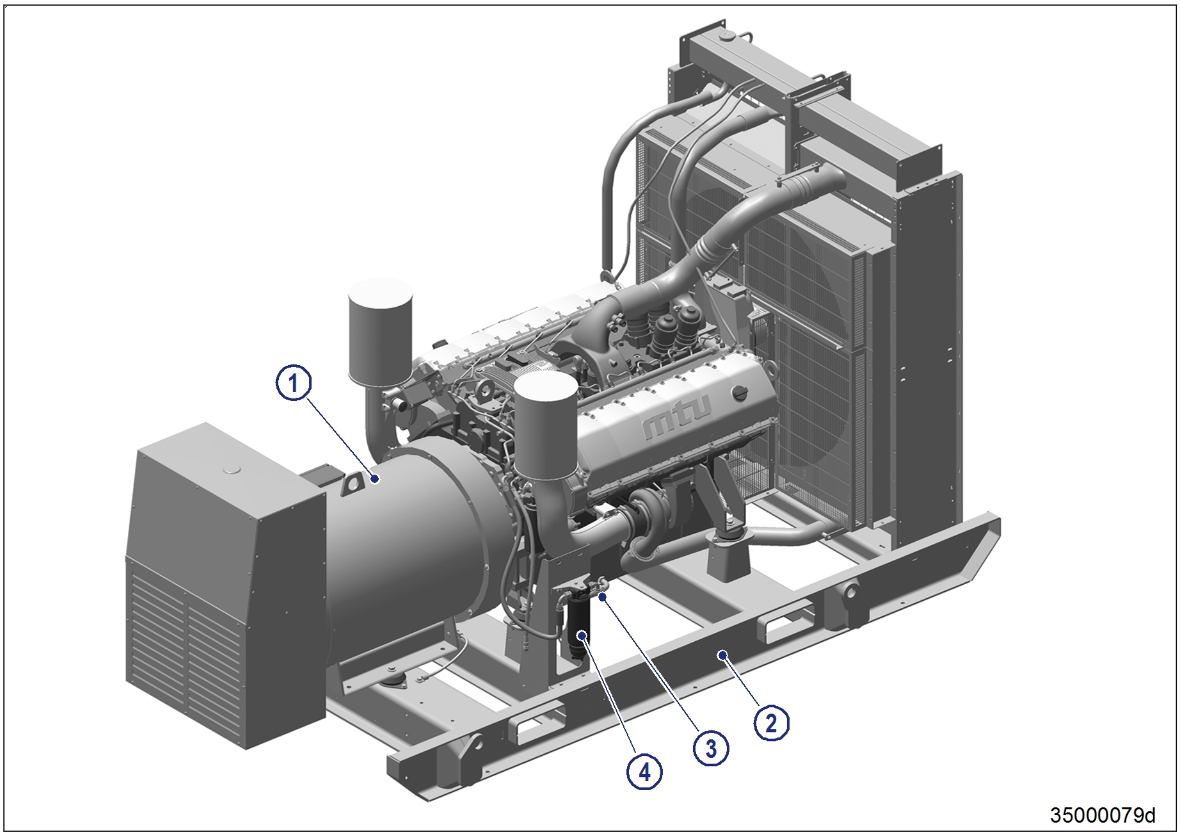

Figure 1: Engine-generator set – Generator side view

1 Generator
2 Base frame
3 Oil drain
4 Fuel prefilter

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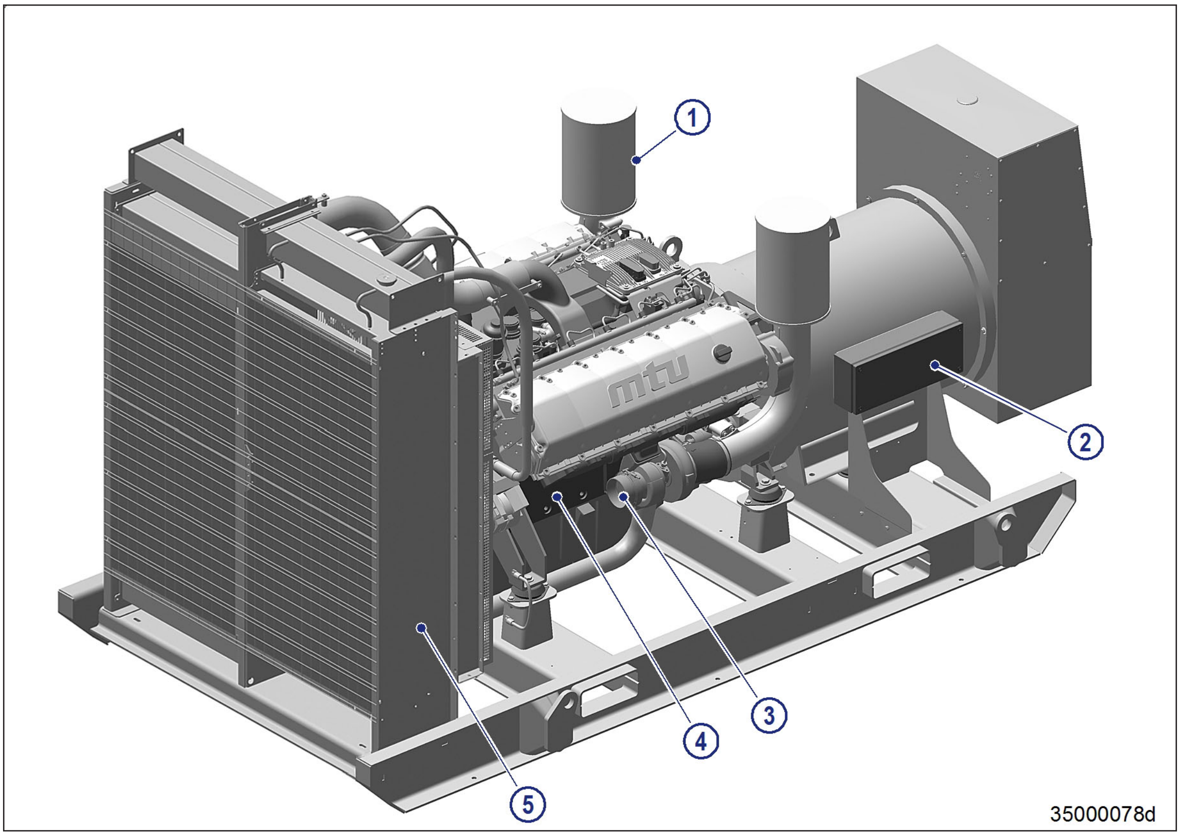

Figure 2: Engine-generator set – Cooler side view

1 Air filter
2 Terminal box
3 Customer exhaust connection
4 Engine
5 Coolant cooler

Standard scope of supply

- MTU Series 10/12 V 1600 Gx0 engine:
  - 1,500 RPM (for 50 Hz)
  - Air-to-air charge-air cooling
  - Electronic engine management system ADEC (ECU8)
- Generator:
  - Outlet box inclusive
  - Voltage 400 V
  - Digital voltage regulator
  - Kit of three CTs for measuring
  - Kit of one CT for paralleling
  - IP-20 protection
- Control version 1: Terminal box without control function (interface between engine-generator set and customer control)
- Smart Connect
- Guards for all rotating parts
- Fuel prefilter with water separation
- Air filter
- Oil drain
- Coolant cooler
- Base frame

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1.3 Engine-generator set – Available options

The figures show an overview of the optional components on an engine-generator set with a 12 V 1600 Gx0 engine. The figures are also valid for an engine-generator set with a 10 V 1600 Gx0 engine.

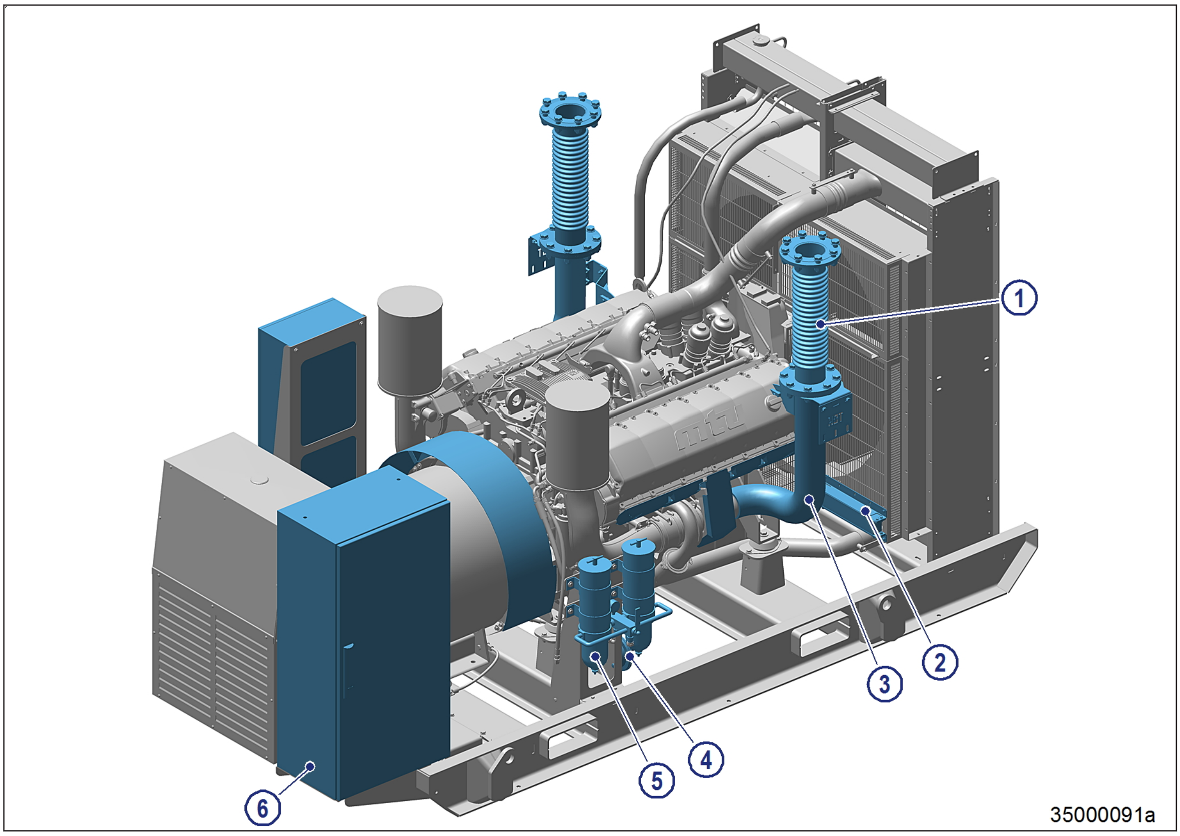

Figure 3: Engine-generator set with options– Generator side view

1 Exhaust bellows
2 Fuel cooler
3 Exhaust elbow
4 Lube oil extraction pump
5 Fuel prefilter with water separator (double)
6 Circuit breaker cabinet

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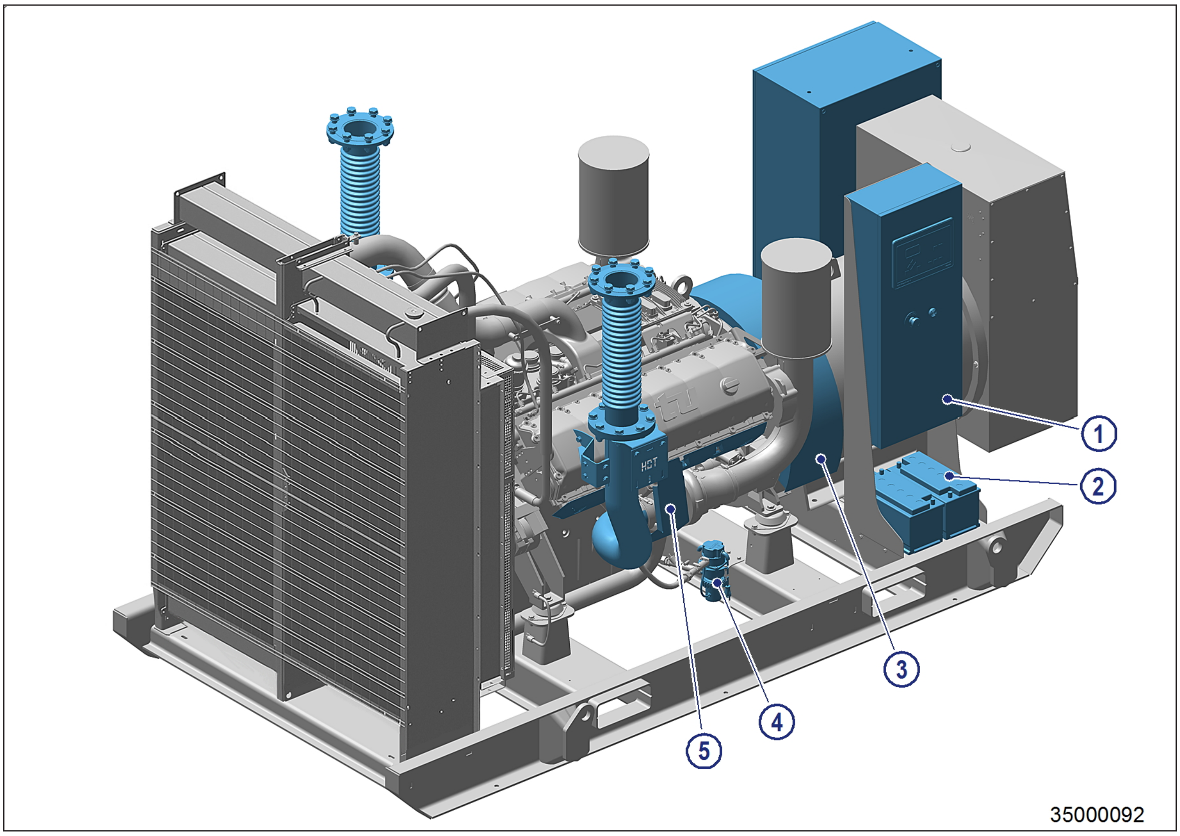

Figure 4: Engine-generator set with options – Cooler side view

1 Control cabinet
2 Starter battery
3 IP-23 drip cover
4 Preheater
5 Heat shield

Available options

- Generator:
  - Voltages 380 V, 415 V
  - Anti-condensation heater
  - IP-23 drip cover
  - Kit of three CTs for differential measuring
- Control version 2, 3a, 3b or 4: Control panel DGC-2020 for management and monitoring
- Fuel prefilter with water separator (single or double)
- Fuel cooler
- Exhaust bellows
- Exhaust elbow
- Heat shield
- Exhaust silencer
- Hand pump for lube oil extraction
- Preheater for engine coolant
- Starter batteries with battery rack and cables
- Battery charger (available in control versions 2, 3a, 3b and 4 as standard, in control version 1 as option)
- Circuit breaker

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2 Engine

2.1 MTU Series 1600 Gx0 engine

The figure shows a 12 V 1600 Gx0 engine. It is also valid for an 8 V or 10 V 1600 Gx0 engine.

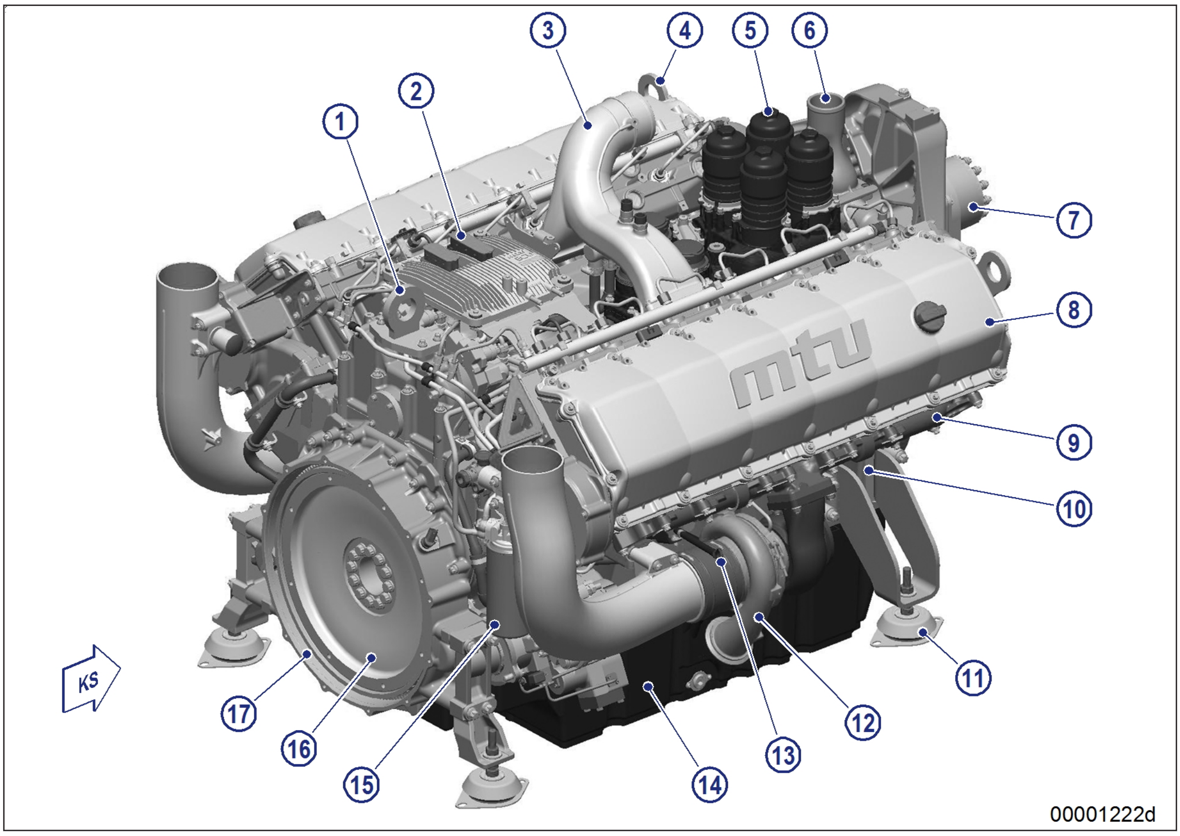

Figure 5: Engine view on driving end

1 Lifting equipment*
2 Engine governor
3 Charge-air intake neck
4 Lifting equipment*
5 Oil filter
6 Thermostat housing
7 Belt drive
8 Cylinder head
9 Exhaust manifold
10 Crankcase
11 Engine mounting
12 Exhaust turbocharger
13 Oil dipstick
14 Oil pan
15 Fuel filter
16 Flywheel
17 Flywheel housing
KS Engine driving end

* Do not use these lifting points to lift the entire engine-generator set.

Series 1600 Gx0 engines with air-to-air charge-air cooling

These engines are compact, powerful, reliable, maintenance-friendly and extremely economical.

Their injection system unites optimum fuel utilization with compliance to the requirements of all relevant environmental protection regulations.

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